Table 3.6

T and μ both constant	f (hertz)	150	300	450
	L (metre)	0.90	0.45	0.30
L and μ both constant	f (hertz)	50	100	150
	T (newton)	60	240	540
L and T both constant	f (hertz)	400	200	100
	μ (kilogram metre^{-1})	0.025	0.10	0.40

9 In investigating the rate P at which heat energy is conducted along a metal rod, a student varies x the length of the rod, and D the temperature difference between its ends. Her seven sets of measurements are shown in Table 3.7.

Table 3.7

Measurement set	x (cm)	D (°C)	P (kJ s^{-1})
1	5	50	0.53
2	5	100	1.05
3	5	150	1.57
4	5	200	2.11
5	10	100	0.51
6	15	100	0.35
7	20	100	0.26

Answer the following *without drawing graphs*.

(*a*) Of the seven sets of measurements, which would you use to investigate the proportionality between P and D?

(*b*) Do the measurements you select suggest a direct or inverse relation between P and D?

(*c*) In order to find the relation between P and D, which of the following would you test for constancy: PD or P/D?

(*d*) Deduce the proportionality existing between P and D.

(*e*) In order to investigate the relationship between P and x, which of the seven sets of readings would you use?

(*f*) Deduce the proportionality between P and x.

10 Water is kept at a constant height in a tank. Water flows from the tank through a number of horizontal tubes of different lengths and cross-sectional areas. These tubes are all attached to the tank at the same level. The volume of water that flows from each of these tubes in 1 minute (the flow rate) is recorded in Table 3.8.

Table 3.8

Length of tube L (cm)	Cross-sectional area of tube S (cm^2)	Flow rate V (cm^3/min)
10	0.010	90
20	0.010	45
30	0.010	30
30	0.020	120
30	0.030	270

(*a*) What graph would be the most useful in predicting the flow rates from tubes of various lengths, all of cross-sectional area 0.010 cm^2, and all at this same level in the tank?

(*b*) What graph would be the most useful in predicting the flow rates from tubes of different cross-sectional areas, all of length 30 cm, and all at this same level in the tank?

(*c*) To what quantity is the flow rate directly proportional?

11 Spheres are allowed to fall vertically through a tank of oil, and are timed between fixed points 1 metre apart. The times of fall for spheres of various radii and densities are given in Table 3.9.

Table 3.9

Density of sphere ρ_s (g cm^{-3})	Radius of sphere r (cm)	Time to fall 1 metre t (s)	$(\rho_s - \rho_0)$ (g cm^{-3})
4	1.00	9	3
4	0.75	16	3
4	0.25	144	3
7	0.25	72	6
10	0.25	48	9

Density of oil, ρ_0, is 1 g cm^{-3}.

(*a*) What graph would be most useful in predicting the time to fall 1 metre for spheres of various radii, all of fixed density 4 g cm^{-3}?

(*b*) What graph would be most useful in predicting the time to fall 1 metre for spheres of various densities, all of fixed radius 0.25 cm?

(*c*) To what quantity is t directly proportional?

12 Four separate flywheels can each be accelerated about its axle by means of a mass m kg, attached to the end of a rope wound around the flywheel, which is allowed to fall.

The number of revolutions through which the wheel turns from rest in 10 s (N_{10}) and in 20 s (N_{20}) are recorded in Table 3.10.

Table 3.10

Wheel	Mass of the wheel M (kg)	Radius of the wheel R (cm)	Number of revs in 10 s N_{10}	Number of revs in 20 s N_{20}
1	2.0	20	20.0	80.0
2	4.0	20	5.0	20.0
3	4.0	10	20.0	80.0
4	8.0	10	5.0	20.0

(*a*) For wheels of the same radius (R), what is the proportionality between N_{10} and M?

(*b*) Let t be the time taken for a flywheel to rotate from rest through a given number of revolutions N. For wheels of the same mass M and radius R, what is the proportionality between N and t?

(*c*) A flywheel has a mass of 1.0 kg and a radius of 20 cm. Starting from rest, through how many revolutions would this flywheel turn in 10 s under the effect of the falling mass of m kg as described above?

(*d*) Starting from rest, through how many revolutions would Wheel 1 turn in 5 s when a mass of m kg is attached to the rope as before?

(*e*) For wheels of the same mass, what is the proportionality between N_{20} and R?

13 In Figure 3.2 graph types B, C, D, F and G all fit the general relation $d = kt^n$, where k and n are constants.

(*a*) In which of the five types would n have the following values?

(*i*) $n > 1$;
(*ii*) $n = 1$;
(*iii*) $1 > n > 0$;
(*iv*) $n = 0$;
(*v*) $n < 0$.

(*b*) From the equation $d = kt^n$, deduce an expression for $\log d$ as a function of $\log t$.

(*c*) If $\log d$ was graphed against $\log t$, then in terms of k and n, what would be

(*i*) the gradient, and
(*ii*) the intercept (also known as the data zero) on the $\log d$ axis?

14 (*a*) From the information given in Figure 3.5, deduce equations expressing

(*i*) $\log_{10} P$ as a function of $\log_{10} V$;
(*ii*) P as a function of V.

(*b*) From Figure 3.6, deduce equations expressing

(*i*) $\log_{10} T$ as a function of $\log_{10} m$;
(*ii*) T as a function of m.

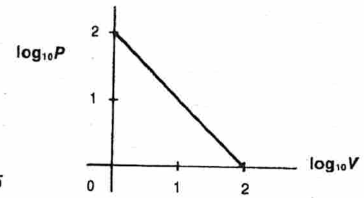

Fig. 3.5

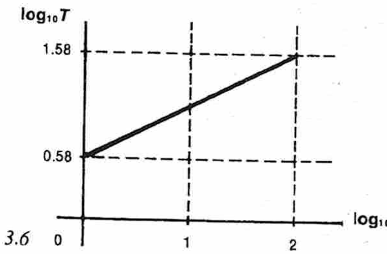

Fig. 3.6

15 An experiment was designed to investigate the dependence of the rate R at which heat energy is given off by a black radiating surface and the thermodynamic temperature T of that surface. R was measured in joule emitted per second from each square metre of the surface, i.e. in watt per square metre. Table 3.11 lists the readings obtained.

Table 3.11

R (W m^{-2})	T (K)
315	273
625	324
1 100	373
2 650	465
6 505	582
17 000	740

Assuming the dependence of R on T is of the form $R = kT^n$, where k and n are constants, draw a graph of $\log_{10} R$ against $\log_{10} T$ to determine the values of k and n (to one significant figure), and deduce an expression for R in terms of T.

16 In an experiment to determine the relation between thermo-e.m.f. and temperature for a thermocouple, a special form of slide wire potentiometer is used to compare the e.m.fs. One junction of the thermocouple is kept at the ice point while the other is immersed in a vessel of water whose temperature is observed with an accurate mercury thermometer. Simultaneous readings of the thermometer and the length of potentiometer wire necessary for balance are listed in Table 3.12.

Table 3.12

Temp (°C)	15.2	27.6	39.7	51.0	62.3	71.9	80.5
Length of wire (cm)	39.7	72.0	102.3	122.1	160.8	184.8	208.0

When balanced against a known e.m.f. of 1.018 millivolt the length of the potentiometer wire is 61.7 cm.

(*a*) Draw a circuit diagram for the arrangement.

(*b*) From a suitably drawn graph, determine a relation connecting the e.m.f. of the thermocouple with the temperature difference between its junctions.

(*c*) What is the melting point of ferrous sulphate if the observed thermo-e.m.f. is 2.74 millivolt?

17 It having been established that the frequency of vibration of a stretched string is inversely proportional to the length of the vibrating segment, the following experiment was performed to investigate the relation between the frequency and the tensile force in the string.

Two wires were stretched across a sounding board. Wire A was adjusted so that the tensile force would remain